

Neuromorphic Computing Chips: Challenges and Trends

Jie Yang, *Member, IEEE*, and Mohamad Sawan, *Life Fellow, IEEE*

CenBRAIN Neurotech, School of Engineering, Westlake University, Hangzhou, Zhejiang, China 310024

Email: yangjie@westlake.edu.cn, sawan@westlake.edu.cn

Abstract— Over the past decade, artificial intelligence (AI) has made unprecedented advancements in various fields. However, with the emergence of large-scale models in recent years, the energy consumption associated with AI computing has become a critical issue that urgently needs to be addressed. Furthermore, as Moore's Law reaches its limits, increasing computational power has become extremely challenging. The scarcity of energy and computational resources presents significant challenges to the further development and application of current AI technologies. These challenges provide an unprecedented opportunity for the introduction of neuromorphic computing chips which show great potential in addressing the problems currently faced by AI. We present in this article the evolution of brain-inspired neuromorphic computing chips, tracing their evolution from the early artificial retina to advanced designs incorporating millions of artificial neurons. Also, we explore future opportunities in key applications such as brain-computer interfaces, which can facilitate more efficient neural communication; embodied intelligence, which seeks to replicate human cognitive functions; and large-scale models, where these chips' energy-efficient processing capabilities can greatly enhance AI system performance and scalability.

Keywords—neuromorphic computing, artificial intelligence, integrated circuits, energy efficiency

I. INTRODUCTION

Over the past decade, artificial intelligence (AI) has made unprecedented advancements across various fields. One of the key driving forces of these advancements is the continuous growth in computational power, which has allowed for the development of more complex and large-scale AI models [1]. Fig. 1 highlights the increasing computational demands of evolving AI models, demonstrating that costs have been doubling every 3.4 months since the advent of deep learning, and accelerating further to double every 2 months with the emergence of large language models (LLMs) [2]. The latest AI models demand computational power capable of processing hundreds of billions of parameters, with performance requirements in the tera- to petaFLOPS range. However, as Moore's Law reaches its physical limits, further advancements in computational power through traditional scaling have become increasingly difficult. Additionally, due to the physical constraints of transistors, enhancing computing efficiency (as shown in Fig. 1) has become a significant challenge [3]. Consequently, the energy requirements for training and running AI models, especially LLMs, are becoming unsustainable, raising concerns about the long-term viability of current AI technologies.

These challenges have created an urgent need for alternative computing paradigms that can offer more energy-efficient solutions. Neuromorphic computing, inspired by the human brain, presents a promising pathway to significantly reduce power consumption while maintaining high levels of performance by mimicking the brain's data processing

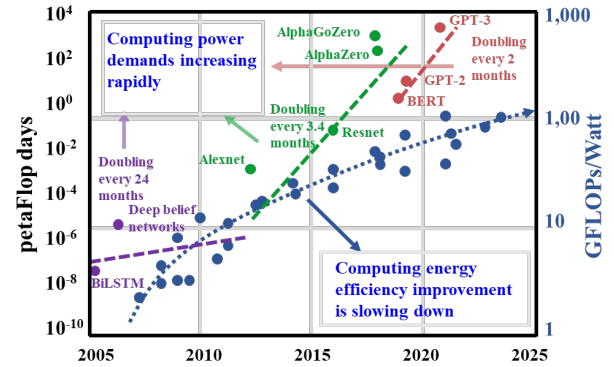


Fig. 1. Computing demands, computing efficiency trends of AI.

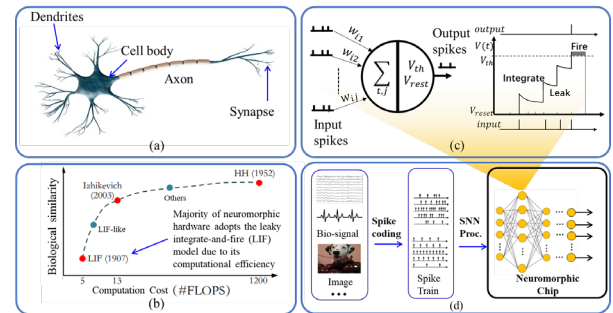


Fig. 2. Neuromorphic computing concept. (a) Biological neuron structure. (b) Different neuron models and computing cost. (c) LIF model. (d) Neuromorphic computing with spiking neural network.

methods. Fig. 2(a) illustrates the basic structure of a biological neuron: dendrites receive electrical signals from other neurons; the cell body (soma) integrates these incoming signals; the axon transmits the electrical impulse in the form of an action potential; and the synapse facilitates communication with the next neuron through chemical or electrical signals. As shown in Fig. 2(b), biological neurons can be mathematically modeled with different computation complexity and accuracy. A simplified model, the leaky integrate-and-fire (LIF) model, is depicted in Fig. 2(c). In this model, the neuron integrates incoming signals over time until a certain threshold is reached, at which point it "fires" an action potential. Neuromorphic chips integrated large scale of such functional neuron models with specific network structures aimed to mimic the data processing of brain (Fig. 2(d)). In neuromorphic chips, signals are represented as binary spikes, eliminating the need for power-intensive multiplication operations. More importantly, signal processing occurs in an event-driven manner, with local memory storage, enabling the real-time processing of large amounts of sensory data while significantly reducing power consumption by avoiding repeated data storage and retrieval. The learning mechanism, spike-timing-dependent plasticity (STDP), allows for the dynamic updating of network parameters based on the timing of neural spikes, unlike traditional backpropagation, which requires error propagation layer by layer. This facilitates efficient on-chip learning.

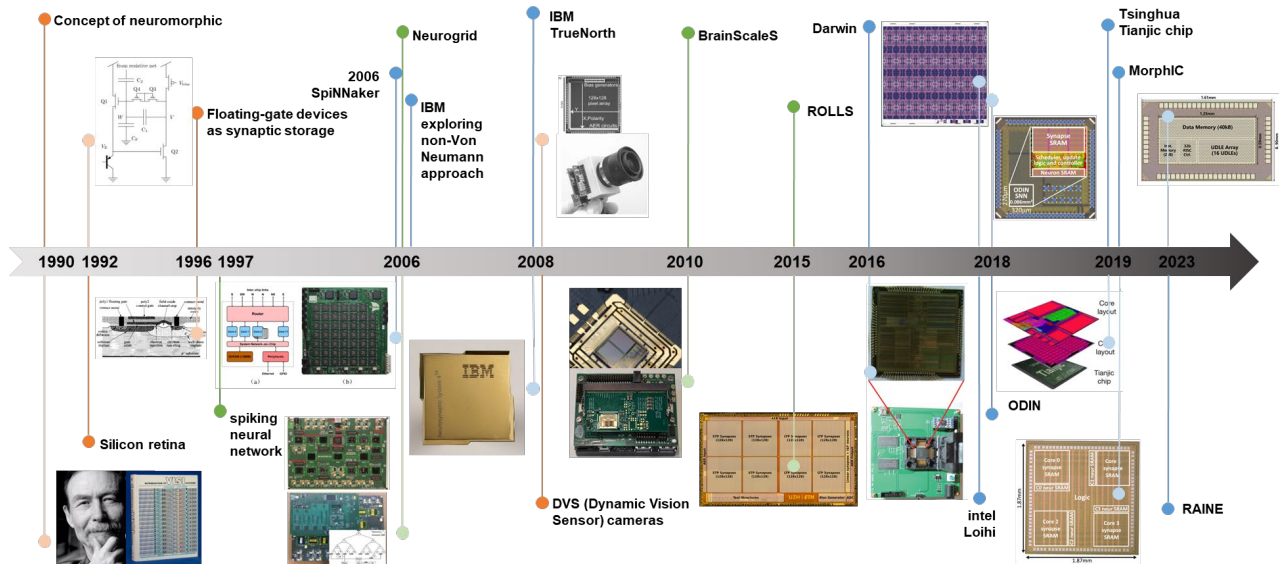


Fig. 3. Evolution of neuromorphic computing chips

Furthermore, system-level extensions, such as Network-on-Chip (NoC) for transmitting neural spike messages, enable scaling up the number of integrated neurons, thereby enhancing the capability of neuromorphic systems to handle complex tasks with low energy requirements.

Despite the significant advantages of neuromorphic computing chips, they are not as widely applied in real-world applications compared to conventional machine learning techniques, primarily due to limitations in accuracy, challenges in training, and development complexities. This paper reviews the evolution of neuromorphic computing chips, tracing their development from early inception to the latest advancements. It evaluates neuromorphic computing from an application perspective and highlights emerging trends and future directions. Future opportunities in key areas such as brain-computer interfaces, embodied intelligence, and large language models (LLMs), where the full potential of neuromorphic chips can be demonstrated, are explored.

II. EVOLUTION OF NEUROMORPHIC COMPUTING CHIPS

Early stage

Fig. 3 illustrates the evolution of neuromorphic computing chips, highlighting milestones in the field. The concept of neuromorphic computing was first proposed by Carver Mead, who observed the similarity between the movement of charges in MOS transistors operated in weak inversion and the flow of ions across the membranes of neurons [4]. Based on Mead's work, Misha Mahowald later advanced this field by using analog electrical circuits to model and mimic the biological functions of rod cells, cone cells, and other excitable cells in the retina [5]. It demonstrates the potential of using transistors to mimic biological functions, paving the way for efficient neural computation. Floating-gate devices were subsequently proposed as non-volatile synaptic storage [6]. The idea of a silicon retina established the foundation for vision chips, where image sensing and processing circuitry are integrated on the same die to mimic human visual system. It allows real-time processing and enhances overall system efficiency. The event-driven feature of human visual system inspires the development of Dynamic Vision Sensor (DVS) cameras [7], now widely used across various fields.

Analog neuromorphic chips

Early neuromorphic computing focused on mimicking the basic data processing capabilities of neurons. In 1997, Wolfgang Maass proposed the formal idea of spiking neural networks (SNNs) [8], introducing a more sophisticated level of processing capabilities to neuromorphic systems. In 2006, the Neurogrid project leveraged the subthreshold electrical properties of silicon transistors to mimic the dynamics of ion channels in neurons [9]. The Neurogrid system comprised neural cores, each containing 256×256 neurons, with 16 neural cores connected in a hierarchical network, allowing the simulation of a network with one million neurons. Similarly, the BrainScaleS project, employed analog circuits to model SNN neuron dynamics, while digital boards equipped with FPGAs managed routing and communication [10]. The system achieved speeds approximately 10,000 times faster than a biological brain, with a power consumption of around 1 kW. The ROLLS (Reconfigurable On-line Learning Spiking Neuromorphic Processor) system, took a different approach, focusing not on large-scale networks but on investigating diverse synaptic plasticity mechanisms, including linear synapses, long-term plasticity (LTP), and short-term plasticity (STP) [11].

Digital neuromorphic chips

In 2006, the SpiNNaker (Spiking Neural Networks architecture) project employed a massively parallel architecture, utilizing ARM processors to implement scalable neural networks and enable efficient real-time neural simulations [12]. Aiming to simulate 1% of the human brain's neural network in real time, this project marked a significant step towards the creation of brain-like hardware systems with commercial digital chips. IBM explored neuromorphic architectures with a non-Von Neumann approach [13]. In 2008, they started developing high-density, low-power, real-time neuromorphic chips, which led to the TrueNorth chip in 2014 [14]. TrueNorth integrated 4,096 cores into a single chip, where each core was capable of simulating up to 256 neurons. These cores were connected through a scalable 2D mesh network, forming a large-scale neuromorphic system capable of handling millions of neurons with minimal power

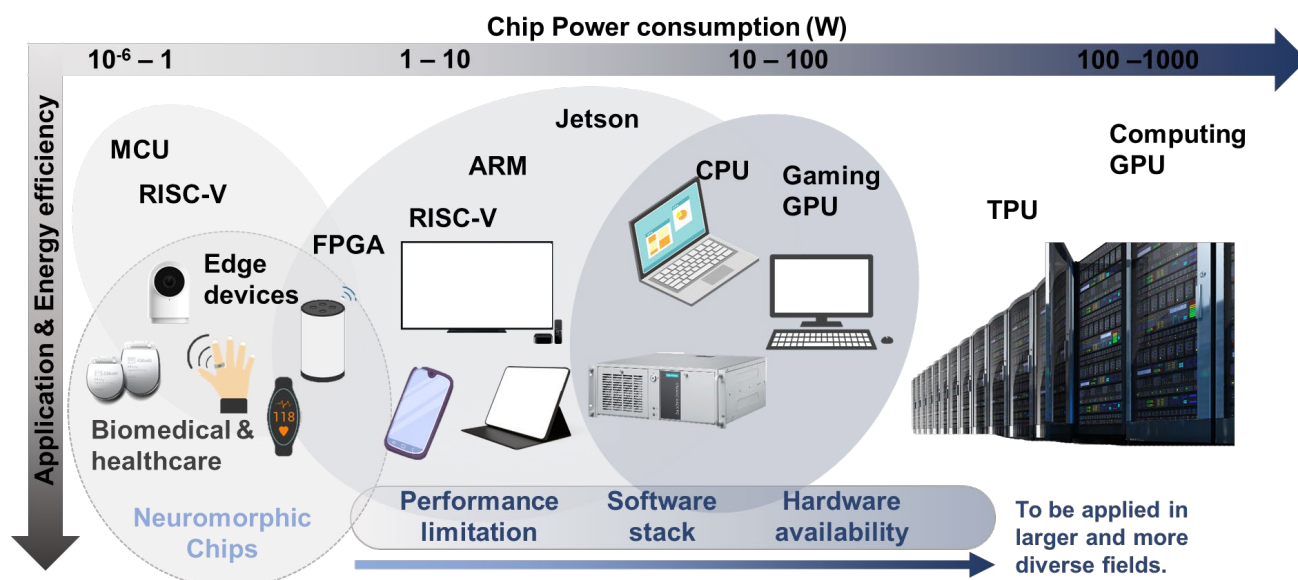


Fig. 4. Neuromorphic applications and unexplored fields.

consumption. Similarly, Intel's Loihi chip integrated 128 neuromorphic cores using digital cores and Network-on-Chip (NoC) interconnections, supporting on-chip learning and demonstrating its capabilities across various neuromorphic applications [15]. In 2019, the Tianjic chip combined traditional AI processing with neuromorphic elements, blending the strengths of both approaches [16]. The Darwin chip, on the other hand, focused on optimizing neuromorphic efficiency [17]. Other projects, such as ODIN [18] and MorphIC [19], RAINE [20] focus on the neuromorphic core design instead of large scale integration have also contributed significantly to the digital neuromorphic research landscape.

III. APPLICATIONS AND LIMITATIONS

Neuromorphic computing algorithms and hardware have been available in the research community for a considerable time, however, the widespread adoption and application of neuromorphic chips have yet to be fully realized. Currently, most research efforts are concentrated on applying neuromorphic computing to edge devices and specialized applications, where stringent constraints on energy consumption, size, and latency are critical. These areas benefit the most from the energy-efficient and real-time processing capabilities of neuromorphic systems. As shown in Fig. 4, neuromorphic computing is applied in edge devices where power consumption ranges from a few microwatts (μ W) to hundreds of milliwatts (mW). For example, edge device for vision [21], audio processing [22], wireless communication [23], human activity recognition [24] etc.

Recently, neuromorphic chips have garnered significant attention in the biomedical and healthcare domain [25], particularly for wearable and implantable devices. For instance, low-power ECG signal processing [26], epilepsy signal detection [27], [28], artificial retina [29], brain-computer interface [30] and neural prosthesis [31].

However, neuromorphic computing has yet to gain sufficient visibility or opportunities to impact larger consumer markets, as illustrated in Fig. 4. In fact, consumer electronics and data centers, where neuromorphic computing seldom appears, are increasingly prioritizing energy efficiency. This need is especially critical in data centers and AI training

facilities, where power consumption is exceptionally high, and improvements in energy efficiency can result in substantial reductions in both operational costs and environmental impact. We argue that there is tremendous potential for neuromorphic computing to make significant contributions in these energy-intensive domains if it is more widely adopted. However, this has not yet been realized due to the following reasons.

Deep learning models, including transformers and their derivatives, have achieved state-of-the-art results in nearly all areas of vision, speech, and language processing. At present, there are no practical applications where neuromorphic computing can clearly surpass these methods in terms of accuracy. This performance gap reduces the motivation to adopt neuromorphic approaches in real-world scenarios, as conventional AI techniques consistently offer higher precision and reliability.

Unlike deep learning, which is supported by a rich ecosystem of tools and accelerated computing hardware like GPUs and CUDA, neuromorphic computing suffers from a lack of well-developed software stacks for training and deployment [32]. This gap in toolsets makes it exceedingly costly and time-consuming to integrate neuromorphic chips into practical applications.

There is a limited number of commercially available neuromorphic hardware options for simulation, training, and deployment [33]. This lack of standardized hardware limits researchers and developers who want to experiment with or implement neuromorphic computing at a scale. The adoption of neuromorphic computing remains constrained.

IV. RECENT TRENDS AND FUTURE DIRECTIONS

The development of deep learning algorithms drives the advancement of dedicated accelerator chips, and this principle similarly applies to neuromorphic computing. The trends in neuromorphic chip development is following the evolution of neuromorphic algorithms. Recognizing the disadvantages when compared with ANNs, the neuromorphic computing community is actively working to bridge the gap. One such approach involves ANN-to-SNN conversion methods, which

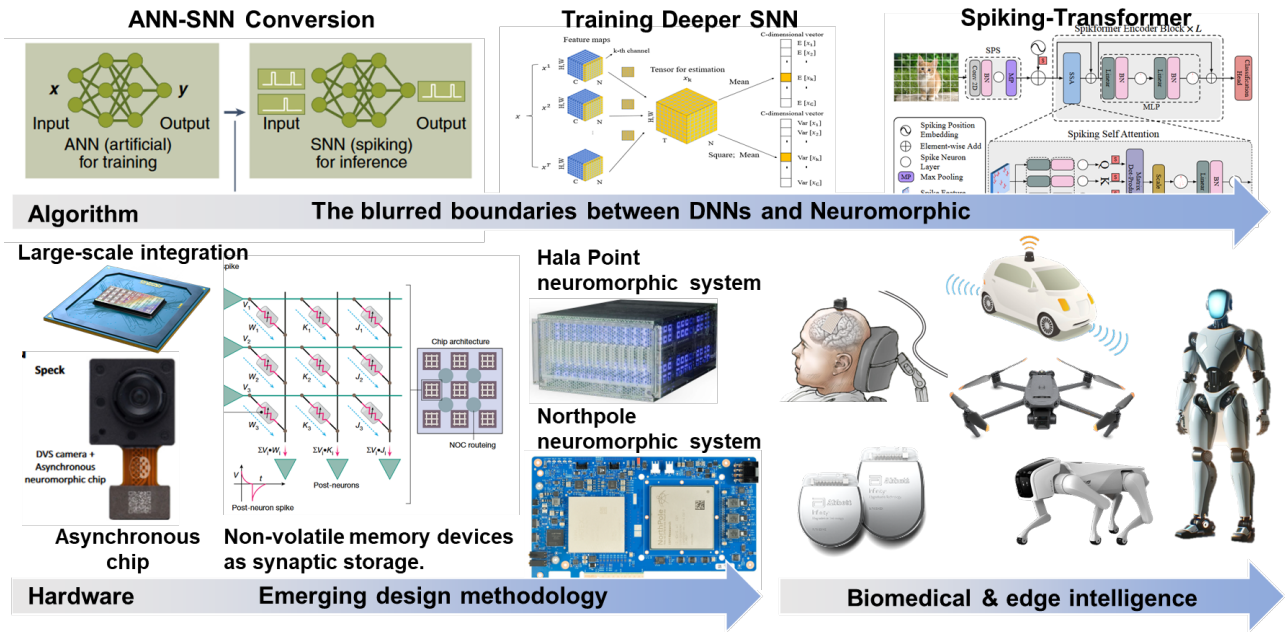


Fig. 5. Trends in neuromorphic algorithms, chips and applications.

equivalently transform ANNs into SNNs. This conversion process replaces the activation functions in ANNs (often ReLU) with spiking neurons in SNNs (commonly integrate-and-fire neurons), while ensuring that the output activation value of the ANN is approximately matched by the spike firing rate in the SNN. The quantization clip-floor-shift function is widely adopted to replace the ReLU activation function in source ANNs [34]. This direct conversion method is enabling neuromorphic computing to rapidly approach state-of-the-art (SOTA) performance across a variety of tasks. On the other hand, emerging techniques threshold-dependent batch normalization [35] have enabled SNNs to reach similar depths as ANN, making the training process more efficient and easier. Furthermore, the transformer architecture has been adapted for the neuromorphic domain [36], [37], significantly improving performance, broadening applications, and further narrowing the gap between neuromorphic computing and traditional ANNs.

The performance and scale of neuromorphic chips are increasing along with the aforementioned neuromorphic computing algorithms. In fact, the advantages of neuromorphic computing are most fully realized in high-power systems, where improvements in energy efficiency yield significant benefits. The gain of neuromorphic computing is evident in large-scale systems such as Intel's Hala point, which incorporates over 1,000 Loihi 2 chips with a billion spiking neurons [38]. Advanced applications such as embodied intelligence and robotics [39], [40], as well as autonomous driving [41] have begun to attract attention within the neuromorphic computing community. However, these large-scale neuromorphic systems still rely on digital integration which mainly operates in a synchronous manner, which somehow diminishes the advantage of neuromorphic computing. Asynchronous digital designs begin to emerge [42], [43], [44] to fully explore the potential of neuromorphic computing, but still have long way to go due to design complexity and lack of library and EDA support. Instead of explore the sparsity using the asynchronous processing, another approach is to utilize the data sparsity of

neuromorphic computing from digital architecture perspective [45], [36].

CONCLUSION

In this paper, we have reviewed the evolution of neuromorphic computing, tracing its development from the early days of analog systems to the latest advancements in digital neuromorphic chips. However, despite various advantages, the application of neuromorphic computing in broader AI domains is still relatively limited. This is primarily due to challenges including lower performance accuracy compared to conventional AI models, a lack of fully developed software ecosystems, and the limited availability of neuromorphic hardware for widespread use. Recent technologies such as ANN-SNN conversion, STBP, threshold-dependent normalization and other training methods are helping to make neuromorphic computing more competitive with existing AI models. Additionally, with large scale integration, neuromorphic systems are entering the field of traditional AI computing, particularly in areas where enhancing power efficiency yields significant benefits. As neuromorphic technology continues to evolve, it is poised to play a larger role in AI, offering a complementary or alternative computing paradigm that could address limitations faced by conventional machine learning methods, especially in the context of large-scale, energy-intensive AI workloads.

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REFERENCES

- [1] Y. LeCun, "1.1 Deep Learning Hardware: Past, Present, and Future," in *2019 IEEE International Solid-State Circuits Conference - (ISSCC)*, San Francisco, CA, USA: IEEE, Feb. 2019, pp. 12–19. doi: 10.1109/ISSCC.2019.8662396.

- [2] A. Mehonic and A. J. Kenyon, "Brain-inspired computing needs a master plan," *Nature*, vol. 604, no. 7905, pp. 255–260, Apr. 2022, doi: 10.1038/s41586-021-04362-w.
- [3] L. Su and S. Naffziger, "1.1 Innovation For the Next Decade of Compute Efficiency," in *2023 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA: IEEE, Feb. 2023, pp. 8–12, doi: 10.1109/ISSCC42615.2023.10067810.
- [4] C. Mead, "Neuromorphic electronic systems," *Proc. IEEE*, vol. 78, no. 10, pp. 1629–1636, Oct. 1990, doi: 10.1109/5.58356.
- [5] M. Mahowald, "VLSI Analogs of Neuronal Visual Processing: A synthesis of form and function".
- [6] C. Diorio, P. Hasler, A. Minch, and C. A. Mead, "A single-transistor silicon synapse," *IEEE Trans. Electron Devices*, vol. 43, no. 11, pp. 1972–1980, Nov. 1996, doi: 10.1109/16.543035.
- [7] P. Lichtsteiner, C. Posch, and T. Delbruck, "A 128x128 120 dB 15 us Latency Asynchronous Temporal Contrast Vision Sensor," *IEEE J. Solid-State Circuits*, vol. 43, no. 2, pp. 566–576, 2008, doi: 10.1109/JSSC.2007.914337.
- [8] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Netw.*, vol. 10, no. 9, pp. 1659–1671, Dec. 1997, doi: 10.1016/S0893-6080(97)00011-7.
- [9] R. Silver, K. Boahen, S. Grillner, N. Kopell, and K. L. Olsen, "Neurotech for Neuroscience: Unifying Concepts, Organizing Principles, and Emerging Tools," *J. Neurosci.*, vol. 27, no. 44, pp. 11807–11819, Oct. 2007, doi: 10.1523/JNEUROSCI.3575-07.2007.
- [10] J. Schemmel, D. Brüderle, A. Gribbl, M. Hock, K. Meier, and S. Millner, "A wafer-scale neuromorphic hardware system for large-scale neural modeling," in *Proceedings of 2010 IEEE International Symposium on Circuits and Systems*, Paris, France: IEEE, May 2010, pp. 1947–1950, doi: 10.1109/ISCAS.2010.5536970.
- [11] N. Qiao *et al.*, "A reconfigurable on-line learning spiking neuromorphic processor comprising 256 neurons and 128K synapses," *Front. Neurosci.*, vol. 9, Apr. 2015, doi: 10.3389/fnins.2015.00141.
- [12] S. B. Furber, F. Galluppi, S. Temple, and L. A. Plana, "The SpiNNaker Project," *Proc. IEEE*, vol. 102, no. 5, pp. 652–665, May 2014, doi: 10.1109/JPROC.2014.2304638.
- [13] P. Merolla, J. Arthur, F. Akopyan, N. Imam, R. Manohar, and D. S. Modha, "A digital neurosynaptic core using embedded crossbar memory with 45pJ per spike in 45nm," in *2011 IEEE Custom Integrated Circuits Conference (CICC)*, San Jose, CA, USA: IEEE, Sep. 2011, pp. 1–4, doi: 10.1109/CICC.2011.6055294.
- [14] P. A. Merolla *et al.*, "A million spiking-neuron integrated circuit with a scalable communication network and interface," *Science*, vol. 345, no. 6197, pp. 668–673, Aug. 2014, doi: 10.1126/science.1254642.
- [15] M. Davies *et al.*, "Loihi: A Neuromorphic Manycore Processor with On-Chip Learning," *IEEE Micro*, vol. 38, no. 1, pp. 82–99, Jan. 2018, doi: 10.1109/MM.2018.112130359.
- [16] J. Pei *et al.*, "Towards artificial general intelligence with hybrid Tianjic chip architecture," *Nature*, vol. 572, no. 7767, pp. 106–111, Aug. 2019, doi: 10.1038/s41586-019-1424-8.
- [17] J. Shen *et al.*, "Darwin: a neuromorphic hardware co-processor based on Spiking Neural Networks," *Sci. China Inf. Sci.*, vol. 59, no. 2, pp. 1–5, Feb. 2016, doi: 10.1007/s11432-015-5511-7.
- [18] C. Frenkel, M. Lefebvre, J.-D. Legat, and D. Bol, "A 0.086-mm² 12.7-pJ/SOP 64k-Synapse 256-Neuron Online-Learning Digital Spiking Neuromorphic Processor in 28nm CMOS," *IEEE Trans. Biomed. Circuits Syst.*, pp. 1–1, 2018, doi: 10.1109/TBCAS.2018.2880425.
- [19] C. Frenkel, J.-D. Legat, and D. Bol, "MorphIC: A 65-nm 738k-Synapse/mm² Quad-Core Binary-Weight Digital Neuromorphic Processor With Stochastic Spike-Driven Online Learning," *IEEE Trans. Biomed. Circuits Syst.*, vol. 13, no. 5, pp. 999–1010, Oct. 2019, doi: 10.1109/TBCAS.2019.2928793.
- [20] C. Fang, C. Wang, S. Zhao, F. Tian, J. Yang, and M. Sawan, "A 510 uW 0.738-mm² 6.2-pJ/SOP Online Learning Multi-Topology SNN Processor With Unified Computation Engine in 40-nm CMOS," *IEEE Trans. Biomed. Circuits Syst.*, vol. 17, no. 3, pp. 507–520, Jun. 2023, doi: 10.1109/TBCAS.2023.3279367.
- [21] H. Yang *et al.*, "Lead facilitated neuromorphic learning for wireless edge artificial intelligence," *Nat. Commun.*, vol. 13, no. 1, p. 4269, Jul. 2022, doi: 10.1038/s41467-022-32020-w.
- [22] A. Jimenez-Fernandez *et al.*, "A Binaural Neuromorphic Auditory Sensor for FPGA: A Spike Signal Processing Approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 4, pp. 804–818, Apr. 2017, doi: 10.1109/TNNLS.2016.2583223.
- [23] N. Skatchkovsky, H. Jang, and O. Simeone, "End-to-End Learning of Neuromorphic Wireless Systems for Low-Power Edge Artificial Intelligence," in *2020 54th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA: IEEE, Nov. 2020, pp. 166–173, doi: 10.1109/IEEECONF51394.2020.9443351.
- [24] V. Fra, E. Forno, R. Pignari, T. C. Stewart, E. Macii, and G. Urgese, "Human activity recognition: suitability of a neuromorphic approach for on-edge AIoT applications," *Neuromorphic Comput. Eng.*, vol. 2, no. 1, p. 014006, Mar. 2022, doi: 10.1088/2634-4386/ac4c38.
- [25] F. Tian, J. Yang, S. Zhao, and M. Sawan, "NeuroCARE: A generic neuromorphic edge computing framework for healthcare applications," *Front. Neurosci.*, vol. 17, p. 1093865, Jan. 2023, doi: 10.3389/fnins.2023.1093865.
- [26] J. Chen, F. Tian, J. Yang, and M. Sawan, "An Event-Driven Compressive Neuromorphic System for Cardiac Arrhythmia Detection," in *2022 IEEE International Symposium on Circuits and Systems (ISCAS)*, Austin, TX, USA: IEEE, May 2022, pp. 2690–2694, doi: 10.1109/ISCAS48785.2022.9937756.
- [27] F. Tian, J. Yang, S. Zhao, and M. Sawan, "A New Neuromorphic Computing Approach for Epileptic Seizure Prediction," in *2021 IEEE International Symposium on Circuits and Systems (ISCAS)*, Daegu, Korea: IEEE, May 2021, pp. 1–5, doi: 10.1109/ISCAS51556.2021.9401560.
- [28] K. Burelo, G. Ramantani, G. Indiveri, and J. Samthein, "A neuromorphic spiking neural network detects epileptic high frequency oscillations in the scalp EEG," *Sci. Rep.*, vol. 12, no. 1, p. 1798, Feb. 2022, doi: 10.1038/s41598-022-05883-8.
- [29] C. Wang, J. Yang, and M. Sawan, "NeuroSEE: A Neuromorphic Energy-Efficient Processing Framework for Visual Prostheses," *IEEE J. Biomed. Health Inform.*, vol. 26, no. 8, pp. 4132–4141, Aug. 2022, doi: 10.1109/JBHI.2022.3172306.
- [30] Z. Liu *et al.*, "Neural signal analysis with memristor arrays towards high-efficiency brain-machine interfaces," *Nat. Commun.*, vol. 11, no. 1, p. 4234, Aug. 2020, doi: 10.1038/s41467-020-18105-4.
- [31] Y. Qi, J. Chen, and Y. Wang, "Neuromorphic computing facilitates deep brain-machine fusion for high-performance neuroprosthesis," *Front. Neurosci.*, vol. 17, p. 1153985, May 2023, doi: 10.3389/fnins.2023.1153985.
- [32] W. Fang *et al.*, "SpikingJelly: An open-source machine learning infrastructure platform for spike-based intelligence," *Sci. Adv.*, vol. 9, no. 40, p. eadi1480, Oct. 2023, doi: 10.1126/sciadv.adi1480.
- [33] S. Bains, "The business of building brains," *Nat. Electron.*, vol. 3, no. 7, pp. 348–351, Jul. 2020, doi: 10.1038/s41928-020-0449-1.
- [34] T. Bu, W. Fang, J. Ding, P. Dai, Z. Yu, and T. Huang, "Optimal ANN-SNN Conversion for High-accuracy and Ultra-low-latency Spiking Neural Networks," Mar. 07, 2023, *arXiv*: arXiv:2303.04347. Accessed: Oct. 09, 2024. [Online]. Available: <http://arxiv.org/abs/2303.04347>
- [35] H. Zheng, Y. Wu, L. Deng, Y. Hu, and G. Li, "Going Deeper With Directly-Trained Larger Spiking Neural Networks," *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 12, pp. 11062–11070, May 2021, doi: 10.1609/aaai.v35i12.17320.
- [36] C. Fang *et al.*, "A 0.078 pJ/SOP Unstructured Sparsity-Aware Spiking Attention/Convolution Processor with 3D Compute Array," in *2024 IEEE Custom Integrated Circuits Conference (CICC)*, Denver, CO, USA: IEEE, Apr. 2024, pp. 1–2, doi: 10.1109/CICC60959.2024.10529019.
- [37] Z. Zhou *et al.*, "Spikformer: When Spiking Neural Network Meets Transformer," Nov. 22, 2022, *arXiv*: arXiv:2209.15425. Accessed: Oct. 09, 2024. [Online]. Available: <http://arxiv.org/abs/2209.15425>
- [38] S. B. Shrestha, J. Timcheck, P. Frady, L. Campos-Macias, and M. Davies, "Efficient Video and Audio Processing with Loihi 2," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Seoul, Korea, Republic of: IEEE, Apr. 2024, pp. 13481–13485, doi: 10.1109/ICASSP48485.2024.10448003.
- [39] Y. Sandamirskaya, M. Kaboli, J. Conradt, and T. Celikel, "Neuromorphic computing hardware and neural architectures for robotics," *Sci. Robot.*, vol. 7, no. 67, p. eabl8419, Jun. 2022, doi: 10.1126/scirobotics.abl8419.
- [40] C. Bartolozzi, G. Indiveri, and E. Donati, "Embodied neuromorphic intelligence," *Nat. Commun.*, vol. 13, no. 1, p. 1024, Feb. 2022, doi: 10.1038/s41467-022-28487-2.
- [41] D. Gehrig and D. Scaramuzza, "Low-latency automotive vision with event cameras," *Nature*, vol. 629, no. 8014, pp. 1034–1040, May 2024, doi: 10.1038/s41586-024-07409-w.
- [42] J. Zhang, H. Wu, J. Wei, S. Wei, and H. Chen, "An Asynchronous Reconfigurable SNN Accelerator With Event-Driven Time Step Update," in *2019 IEEE Asian Solid-State Circuits Conference (A-SSCC)*, Macao, Macao: IEEE, Nov. 2019, pp. 213–216, doi: 10.1109/A-SSCC47793.2019.9056903.
- [43] J. Zhang *et al.*, "22.6 ANP-I: A 28nm 1.5pJ/SOP Asynchronous Spiking Neural Network Processor Enabling Sub-0.1 μJ/Sample On-Chip Learning for Edge-AI Applications," in *2023 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA: IEEE, Feb. 2023, pp. 21–23, doi: 10.1109/ISSCC42615.2023.10067650.
- [44] M. Yao *et al.*, "Spike-based dynamic computing with asynchronous sensing-computing neuromorphic chip," *Nat. Commun.*, vol. 15, no. 1, p. 4464, May 2024, doi: 10.1038/s41467-024-47811-6.
- [45] G. K. Chen, R. Kumar, H. E. Sumbul, P. C. Knag, and R. K. Krishnamurthy, "A 4096-Neuron 1M-Synapse 3.8-pJ/SOP Spiking Neural Network With On-Chip STDP Learning and Sparse Weights in 10-nm FinFET CMOS," *IEEE J. Solid-State Circuits*, vol. 54, no. 4, pp. 992–1002, Apr. 2019, doi: 10.1109/JSSC.2018.2884901.